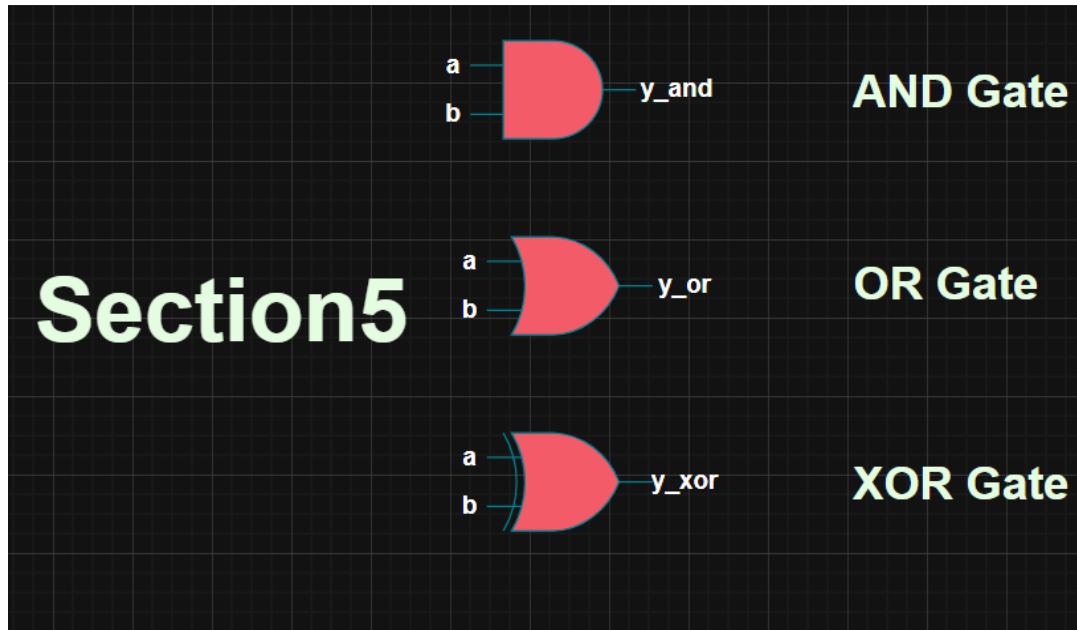


NAME: MUHAMMAD USMAN

REG. NO# 2024-EE-171

1-EDA Playground link: <https://edaplayground.com/x/s7B7>

2-Schematic Diagram:



Section5.draw.io

3-Output Screenshot:

```
Log Share
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 16 08:32 2026
a=0, b=0, y_and=0, y_or=0, y_xor=0
a=0, b=1, y_and=0, y_or=1, y_xor=1
a=1, b=0, y_and=0, y_or=1, y_xor=1
a=1, b=1, y_and=1, y_or=1, y_xor=0
$finish called from file "testbench.sv", line 26.
$finish at simulation time 40
V C S S i m u l a t i o n R e p o r t
Time: 40 ns
CPU Time: 0.530 seconds; Data structure size: 0.0Mb
Thu Jul 16 08:32:05 2026
Finding VCD file...
./dump.vcd
[2026-07-16 12:32:07 UTC] Opening EPWave...
Done
```

Task1_log_result.png

Log

Share

Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 16 09:02 2026
[PASS]: a=0, b=0, AND=0, OR=0, XOR= 0
[PASS]: a=0, b=1, AND=0, OR=1, XOR= 1
[PASS]: a=1, b=0, AND=0, OR=1, XOR= 1
[PASS]: a=1, b=1, AND=1, OR=1, XOR= 0
\$finish called from file "testbench.sv", line 49.
\$finish at simulation time 40
V C S S i m u l a t i o n R e p o r t
Time: 40 ns
CPU Time: 0.550 seconds; Data structure size: 0.0Mb
Thu Jul 16 09:02:04 2026
Finding VCD file...
./dump.vcd
[2026-07-16 13:02:04 UTC] Opening EPWave...
Done

Task2_log_result.png

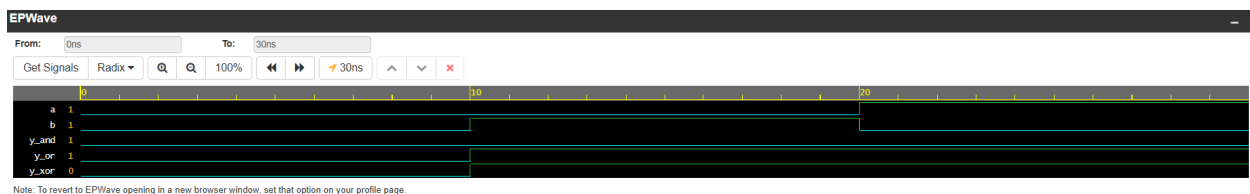
Log

Share

Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64; Jul 16 09:03 2026
[PASS]: a=0, b=0, AND=0, OR=0, XOR= 0
[FAIL]: a=0, b=1, AND=0, OR=1, XOR= 1
[FAIL]: a=1, b=0, AND=0, OR=1, XOR= 1
[PASS]: a=1, b=1, AND=1, OR=1, XOR= 0
\$finish called from file "testbench.sv", line 49.
\$finish at simulation time 40
V C S S i m u l a t i o n R e p o r t
Time: 40 ns
CPU Time: 0.490 seconds; Data structure size: 0.0Mb
Thu Jul 16 09:03:49 2026
Finding VCD file...
./dump.vcd
[2026-07-16 13:03:49 UTC] Opening EPWave...
Done

Intentional_error_log_result.png

4-Waveform Screenshot:



Waveform.png

Extra:

```
// expected results  
exp_and = a & b;  
exp_or  = a | b;  
exp_xor = 0;
```

intentional_error.png